Discussion of "Deep Fréchet Regression"

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We congratulate Iao, Zhou, and Müller on a creative and timely contribution to regression with object-valued responses. Classical regression starts with predictors $\boldsymbol{X} \in \mathbb{R}^p$ and a response Y in $\mathbb{R}$ or $\mathbb{R}^q$. In Iao et al. (2025), the response instead belongs to a metric space (Ω, d) and may be a probability distribution, network, covariance matrix, tree, or another non-Euclidean object. The regression target is the conditional Fréchet mean $m(\boldsymbol{x}) = E_{\oplus}(Y \mid \boldsymbol{X} = \boldsymbol{x}) := \arg\min_{y \in \Omega} \mathrm{E}\{d^2(Y, y) \mid \boldsymbol{X} = \boldsymbol{x}\}$. The challenge is to accommodate multivariate or high-dimensional predictors without imposing linear structure on Ω.

Deep Fréchet regression (DFR) in this paper addresses this challenge through a modular three-stage construction. First, ISOMAP (Tenenbaum et al. 2000) gives Euclidean coordinates $\boldsymbol{Z}_i = \widehat{\boldsymbol{\psi}}(Y_i) \in \mathbb{R}^r$ for the observed responses. Second, deep neural networks (DNNs) predict these coordinates from $\boldsymbol{X}$. Third, multivariate local Fréchet regression (Petersen & Müller 2019), LFR, maps the predicted coordinates back to Ω. In compact notation, $Y_i \xrightarrow{\widehat{\psi}} \boldsymbol{Z}_i$, $\boldsymbol{X} \xrightarrow{\widehat{\boldsymbol{g}}} \widehat{\boldsymbol{Z}}$, and $\widehat{\boldsymbol{Z}} \xrightarrow{\widehat{v}_h} \widehat{m}(\boldsymbol{X}) = \widehat{v}_h(\widehat{\boldsymbol{g}}(\boldsymbol{X}))$. DFR's modular construction retains access to the original response metric, returns predictions to the object space, and yields a pipeline-level convergence rate conditional on an explicit representation-error model.

We consider three questions: (i) which manifold is learned from the responses, (ii) whether representation error can be derived from geometric and sampling conditions, and (iii) how the estimator depends on the latent coordinates. All concern the response geometry that DFR must estimate.

1 Which manifold is being learned?

The conditional Fréchet mean is defined by minimization over the full response space Ω. The paper then imposes the additional structural assumption that the resulting minimizer lies on a low-dimensional manifold: $m(\boldsymbol{x}) \in \mathcal{M} \subset \Omega$, $\boldsymbol{\psi} : \mathcal{M} \longrightarrow \mathbb{R}^r$. Let $\mathcal{X} = \text{supp}(\boldsymbol{X})$ denote the predictor support. It is useful to distinguish the postulated manifold $\mathcal{M}$ from the closed regression image

$$\mathcal{M}_0 = \overline{\{m(\boldsymbol{x}) : \boldsymbol{x} \in \mathcal{X}\}}. \tag{1}$$

We henceforth assume that $\mathcal{M}$ is closed, so that $\mathcal{M}_0 \subseteq \mathcal{M}$. Recovering all of $\mathcal{M}$ requires $\mathcal{M}_0 = \mathcal{M}$, together with sufficient design coverage. Otherwise, only the regression image $\mathcal{M}_0$ is identifiable from the regression relationship.

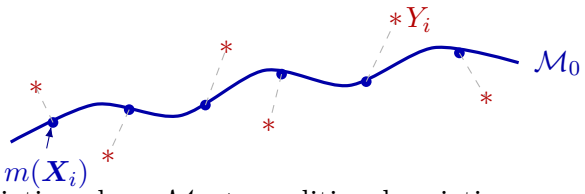


Figure 1: *The regression image $\mathcal{M}_0$ is traced by conditional Fréchet means, whereas* ISOMAP *uses the noisy responses Y_i, which may lie away from $\mathcal{M}_0$.*

ISOMAP uses the observed responses $\{Y_i\}_{i=1}^n$, not the conditional means $\{m(\boldsymbol{X}_i)\}_{i=1}^n$. Because Y_i need not lie on $\mathcal{M}$, the response cloud combines movement of $m(\boldsymbol{X}_i)$ along $\mathcal{M}_0$ with conditional variation around it (Figure 1). Large, heteroscedastic, or anisotropic variation can alter nearest neighbors and graph shortest paths.

A simple metric inequality makes the issue transparent. Let $m_i = m(\boldsymbol{X}_i)$. For $1 \le i, j \le n$, the triangle inequality gives $|d(Y_i, Y_j) - d(m_i, m_j)| \le d(Y_i, m_i) + d(Y_j, m_j)$. Define $\rho_n = \max_{1 \le i \le n} d(Y_i, m_i)$. Then $\max_{1 \le i,j \le n} |d(Y_i, Y_j) - d(m_i, m_j)| \le 2\rho_n$. Consequently, a strong sufficient route to stability of the neighborhood graph would make ρ_n small relative to the

ISOMAP neighborhood scale, the relevant separation between points, and the geometric scale set by curvature or reach. With one response per predictor value and nonvanishing conditional noise, however, ρ_n generally need not tend to zero, so this uniform route is a strong sufficient condition rather than a generic consequence of the model.

This observation motivates a closer look at the representation discrepancy $Z_{ij} - Z_{ij}^0$. Condition (C2) in Iao et al. (2025) assumes, for each coordinate j, that

$$Z_{ij} - Z_{ij}^0 = \pi_j(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n)(\epsilon_{ij} + u_{nj}), \qquad \boldsymbol{Z}_i = \widehat{\boldsymbol{\psi}}(Y_i), \quad \boldsymbol{Z}_i^0 = \boldsymbol{\psi}(m(\boldsymbol{X}_i)), \tag{2}$$

where $u_{nj} \to 0$ and ϵ_{ij} is mean-zero sub-Gaussian noise. Condition (2) does *not* require $\boldsymbol{Z}_i - \boldsymbol{Z}_i^0 \to 0$ for each observation. Rather, because ϵ_{ij} is independent of the predictors, it yields the conditional alignment

$$\mathrm{E}(Z_{ij} \mid \boldsymbol{X}_1, \ldots, \boldsymbol{X}_n) - Z_{ij}^0 = \pi_j(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n) u_{nj} \longrightarrow 0, \tag{3}$$

provided π_j remains bounded. The DNN stage can average the remaining random coordinate error.

Equation (2) also imposes more structure than conditional centering: for each j, given $\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n$, the centered errors are independent and share the common scale $|\pi_j(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n)|$. Yet ISOMAP is a global function of all responses, so its coordinate errors may remain conditionally dependent, while heteroscedastic $Y \mid \boldsymbol{X}$ may require observation-specific scales. It would therefore be useful to replace the exact factorization by vector-level concentration and dependence conditions, or to allow scales $\pi_{ij}(\boldsymbol{X}_1, \ldots, \boldsymbol{X}_n)$.

Accordingly, the key geometric question is not whether every observed response is mapped close to its conditional mean, but when unsupervised ISOMAP coordinates of the response cloud have conditional centers aligned with coordinates of the regression image $\mathcal{M}_0$.

This distinction seems especially consequential when the conditional spread changes with $\boldsymbol{X}$, has a strong tangential component, or is asymmetric in the ambient geometry. In such

cases, even the conditional center of an embedded response may not equal the embedding of its conditional Fréchet mean. It would be valuable to identify primitive conditions on the conditional law $Y \mid \boldsymbol{X}$, on the coverage of $\mathcal{M}_0$, and on the localization parameter of ISOMAP that imply (3) and the more specific error structure in (2).

2 From representation error to manifold fitting

Condition (C2) carries representation uncertainty through the DNN and LFR stages, but bundles four sources of error: response variation around $m(\boldsymbol{X}_i)$, perturbation of neighborhoods and graph distances, finite-sample embedding error and distortion, and bias from fitting the marginal response cloud rather than $\mathcal{M}_0$.

When the response space has a suitable ambient structure, $\mathcal{M}_0$ is sufficiently regular to admit a well-defined local projection $P_{\mathcal{M}_0}$, defined by $P_{\mathcal{M}_0}(y) = \arg\min_{z \in \mathcal{M}_0} d(y, z)$. If $\widehat{\boldsymbol{\psi}}$ has been aligned with $\boldsymbol{\psi}$, one possible decomposition is

$$\widehat{\boldsymbol{\psi}}(Y_i) - \boldsymbol{\psi}(m(\boldsymbol{X}_i)) = \underbrace{\widehat{\boldsymbol{\psi}}(Y_i) - \boldsymbol{\psi}(P_{\mathcal{M}_0}(Y_i))}_{\text{off-manifold and embedding error}} + \underbrace{\boldsymbol{\psi}(P_{\mathcal{M}_0}(Y_i)) - \boldsymbol{\psi}(m(\boldsymbol{X}_i))}_{\text{tangential response variation}}.$$

The first term reflects two sources of ISOMAP error: conditional variation away from $\mathcal{M}_0$ can perturb the neighborhood graph and its shortest-path distances, while the subsequent multidimensional-scaling step may represent those graph distances only approximately in $\mathbb{R}^r$. The preceding alignment is necessary because ISOMAP coordinates are identifiable only up to a rigid transformation. For example, a natural global error is

$$\Delta_n = \inf_{R \in \mathcal{O}(r),\, \boldsymbol{b} \in \mathbb{R}^r} \left\{ \frac{1}{n} \sum_{i=1}^{n} \left\| R\boldsymbol{Z}_i + \boldsymbol{b} - \boldsymbol{Z}_i^0 \right\|^2 \right\}^{1/2}, \tag{4}$$

where $\mathcal{O}(r) = \{R \in \mathbb{R}^{r \times r} : R^\top R = \mathbf{I}_r\}$ denotes the orthogonal group. This quantity separates the scientifically relevant representation error from the arbitrary rotation, reflection, and translation of an embedding.

In the special case where $\Omega \subseteq \mathbb{R}^D$ and d is induced by the Euclidean norm, statistical manifold-estimation theory offers useful templates for turning this decomposition into rates. Existing theory quantifies manifold recovery in Hausdorff loss under noisy sampling (Genovese et al. 2012) and gives nonasymptotic rates for estimating a manifold, its tangent spaces, and its curvature under regularity conditions involving reach and sampling density (Aamari & Levrard 2019). These results do not apply automatically to general object spaces or to the regression image $\mathcal{M}_0$ in (1); here the points on the target manifold are latent and their sampling distribution is induced by $\boldsymbol{X}$. This also clarifies a distinction central to manifold fitting: dimension reduction returns latent coordinates, whereas manifold fitting aims to reconstruct the geometric object itself, together with quantities such as its tangent spaces or curvature; Yao & Xia (2025) develop one such formulation under unbounded noise. In DFR, that object is the regression image $\mathcal{M}_0$, not necessarily the marginal response cloud. Nevertheless, these Euclidean manifold-estimation results suggest a route toward bounds of the schematic form $\Delta_n \lesssim \mathcal{E}_{\text{noise}} + \mathcal{E}_{\text{graph}} + \mathcal{E}_{\text{embed}}$, where the terms depend on conditional noise magnitude, reach or curvature, sampling fill distance, the ISOMAP neighborhood size, target dimension, and Euclidean embedding stress. In a general metric space, Hausdorff or Gromov–Hausdorff analogues and local metric-entropy conditions may be more appropriate than ambient reach.

Because Δ_n also includes the stochastic coordinate error induced by response variation, it need not converge to zero under nonvanishing conditional noise. For a direct connection with Condition (C2), one would additionally need to control the aligned conditional bias and characterize the scale and dependence of the centered representation errors.

Such a result would connect directly to the paper's main DFR rate. Theorem 2 has the form $d(\widehat{m}(\boldsymbol{X}), m(\boldsymbol{X})) = O_p(h^{2/(\gamma_1-1)} + (nh^r)^{-1/\{2(\gamma_2-1)\}} + (h^{-r-1}\zeta_n)^{1/(\gamma_3-1)})$, where $\zeta_n = \kappa_n \log^{3/2} n + u_n$. The first two terms are the smoothing bias and stochastic error of

multivariate local Fréchet regression. The third transports representation and DNN errors through the errors-in-variables step. Deriving explicit geometric bounds for the bias u_n, together with the scale and dependence of the centered representation errors, would make the full error budget interpretable in terms of noise, curvature, sampling density, and tuning parameters.

The regression setting also suggests ways to exploit the predictors when learning the manifold. For example, one might apply ISOMAP to cross-fitted pilot estimates of $m(\boldsymbol{X}_i)$ rather than directly to Y_i; alternate between estimating m and fitting $\mathcal{M}_0$; or regularize an embedding so that nearby predictor values have compatible latent conditional centers. These approaches trade the clean modularity of DFR for a closer match between the manifold used by the algorithm and the regression manifold appearing in the model. They also motivate practical diagnostics: stability across intrinsic dimensions and neighborhood sizes, subsample stability after Procrustes alignment, and comparisons of embeddings based on raw versus locally denoised responses.

3 Global representation and coordinate dependence

The assumed map $\boldsymbol{\psi} : \mathcal{M} \to \mathbb{R}^r$ deserves a topological qualification. If it is a continuous one-to-one global chart, it is stronger than $\dim(\mathcal{M}) = r$: a circle, sphere, or torus requires a cut, multiple charts, or an embedding into $\mathbb{R}^s$, $s > r$.

Even when a global representation exists, it defines the DNN loss, the neighborhoods and kernel weights used by local Fréchet regression, and the bandwidth scale. Coordinate choice can therefore affect finite-sample object-valued predictions.

Let $v : \mathbb{R}^r \to \Omega$ satisfy $m = v \circ \boldsymbol{g}_0$. For a bijection $T : \mathbb{R}^r \to \mathbb{R}^r$, set $\boldsymbol{g}_T = T \circ \boldsymbol{g}_0$ and $v_T = v \circ T^{-1}$. Then $v_T(\boldsymbol{g}_T(\boldsymbol{x})) = m(\boldsymbol{x})$ at the population level, but finite-sample DFR is

less invariant because its two stages are estimated separately.

Rigid motions preserve Euclidean distances and radial-kernel weights, although exact equivariance also requires compatible DNN training. Anisotropic or nonlinear reparameterizations change neighborhoods, bandwidth geometry, and the approximation complexity of $T \circ \boldsymbol{g}_0$. Bi-Lipschitz regularity may preserve consistency, but not finite-sample estimates or constants.

This suggests using aligned-distance errors, bandwidths adapted to an estimated pullback metric, or LFR weights based on estimated intrinsic distances. At a minimum, sensitivity to the embedding algorithm, intrinsic dimension, and coordinate normalization should accompany the usual DNN and bandwidth tuning.

4 Concluding perspective

DFR makes an important contribution by showing how manifold learning, deep neural networks, and local Fréchet regression can be assembled into a flexible and theoretically analyzable procedure for metric-space-valued responses.

The three questions raised in Sections 1–3 are forward-looking and build on the paper's main contribution. They point from regression on a learned representation toward regression with a statistically estimated response geometry. The authors' subsequent end-to-end metric regression framework (Zhou et al. 2026) provides a complementary direction that avoids an explicit latent manifold. The modular DFR construction nevertheless has distinctive advantages in interpretability and error decomposition. A promising hybrid would preserve these geometric advantages while allowing the representation to be informed by prediction loss in the original metric space. In this formulation, geometry is not merely a device for dimension reduction; it becomes part of the signal to be estimated (Meng 2026).

References

Aamari, E. & Levrard, C. (2019), 'Nonasymptotic rates for manifold, tangent space and curvature estimation', *Annals of Statistics* **47**(1), 177–204.

Genovese, C. R., Perone-Pacifico, M., Verdinelli, I. & Wasserman, L. (2012), 'Manifold estimation and singular deconvolution under Hausdorff loss', *Annals of Statistics* **40**(2), 941–963.

Iao, S. I., Zhou, Y. & Müller, H.-G. (2025), 'Deep Fréchet regression', *Journal of the American Statistical Association* **120**(551), 1437–1448.

Meng, X.-L. (2026), 'Of shapes and curves—why AI should use more Chernoff faces, and data science more manifold fitting', *Harvard Data Science Review* **8**(3).

Petersen, A. & Müller, H.-G. (2019), 'Fréchet regression for random objects with Euclidean predictors', *Annals of Statistics* **47**(2), 691–719.

Tenenbaum, J. B., de Silva, V. & Langford, J. C. (2000), 'A global geometric framework for nonlinear dimensionality reduction', *Science* **290**(5500), 2319–2323.

Yao, Z. & Xia, Y. (2025), 'Manifold fitting under unbounded noise', *Journal of Machine Learning Research* **26**(45), 1–55.

Zhou, Y., Iao, S. I. & Müller, H.-G. (2026), 'End-to-end deep learning for predicting metric space-valued outputs', *Journal of Machine Learning Research* **27**(132), 1–38.